Let C_0 be a baseline copula (empirical or parametric) calibrated from longitudinal data. C_0 provides a *dependence template* for how students move from U to V and induces the conditional CDF (transition kernel)

$$F_0(v \mid u) \;=\; \Pr(V \le v \mid U = u) \;=\; \frac{\partial}{\partial u} C_0(u, v).$$

Its inverse in v, the conditional quantile function/kernel, is

$$Q_0(p \mid u) = F_0^{-1}(p \mid u).$$

Intuitively, for each starting quantile u, the kernel $F_0(\cdot \mid u)$ describes the distribution of possible ending quantiles V under the baseline dependence structure C_0.

Let $P \in [0, 1]$ be a random variable (i.e., a latent SGPc on the 0–1 scale) and p denote a realization in $[0, 1]$. A *growth regime*, H, is the distribution of P represented by its CDF:

$$H(p) \;:=\; \Pr(P \le p), \qquad p \in [0, 1].$$

Given $U = u$ and a draw $P = p$, the model generates an outcome percentile by[1]

$$V \;=\; Q_0(P \mid U = u).$$

A growth regime governs how the subgroup S "occupies" the baseline kernel $F_0(\cdot \mid u)$. The baseline growth regime is Uniform(0, 1) and its CDF is the identity on $[0, 1]$. That is,

$$P \sim \text{Uniform}(0, 1) \quad \Longleftrightarrow \quad H(p) = p.$$

Departures from $H(p) = p$ tilt mass toward lower or higher conditional percentiles relative to the uniform baseline regime. Although we do not observe pairs (u_i, v_i) in TIMSS- or NAEP-style settings, the model provides a *forward map* from the prior to current margin.

Thus, for any fixed u,

$$\begin{aligned}
\Pr(V \le v \mid U = u) \;&=\; \Pr\big(Q_0(P \mid u) \le v\big) \\
&=\; \Pr\big(P \le F_0(v \mid u)\big) \\
&=\; H\big(F_0(v \mid u)\big).
\end{aligned}$$

Therefore, by the law of total probability, the subgroup's predicted current-grade marginal CDF is

$$\begin{aligned}
F_H(v) \;&=\; \Pr(V \le v) \\
&=\; \mathbb{E}_U\left[H\big(F_0(v \mid U)\big)\right] \\
&=\; \int_0^1 H\big(F_0(v \mid u)\big)\, dF_U(u) \\
&\approx\; \frac{\sum_{i=1}^{n} w_i\, H\big(F_0(v \mid u_i)\big)}{\sum_{i=1}^{n} w_i},
\end{aligned}$$

where $\{u_i, w_i\}$ are the observed prior pseudo-observations and survey weights for the subgroup (i.e., a weighted empirical approximation to F_U).